

Chapter 1

Understanding Cloud Computing

1.1 Introduction to Cloud Computing

1.1.1 Introduction to cloud computing

1.1.2 The power of cloud

- Vertical and Horizontal Scaling
- Pay as you go pricing
- Speed
- Reliability
- Security

1.1.3 Cloud service models

- IaaS: owning car
- PaaS: ridesharing
- SaaS: taking a bus
- FaaS: Function as a Service, auth and transaction
- XaaS: Anything as a Service, covers everything

1.2 Cloud Deployment

1.2.1 Cloud Deployment models

- Private cloud: uses virtualization for on demand resources and can be located off-premises, direct control over resources available, and how data is stored
- Public cloud: internet accessible for general public, no access to data center and hardware
- Hybrid cloud: mix, for seasonal spikes or cloud bursting
- Multicloud : combine different CSP
- Community cloud: shared by specific community with common interest

1.2.2 Regulations on the cloud

Different regions have different views on how data should be managed

- General Data Protection Regulation (GDPR): EU, explicit consent, data breaches notification, encrypted, anonymized PI, can't leave EU borders unless guarantee same level of protection, fine of 20 mil or 4% of worldwide revenue whichever is greater
- https://commission.europa.eu/law/law-topic/data-protection/data-protection-explained_en
- Brazil's Lei Geral de Proteção de Dados (LGPD)
- California's Consumer Privacy Act (CCPA)
- USA's Health Insurance Portability and Accountability Act (HIPAA)
- Japan's Act on Protection of Personal Information
- Thailand Personal Data Protection Act (PDPA)
- Canada's Personal Information and Electronic Documents Act (PIPEDA)

1.2.3 Cloud computing roles

- Data scientist: run computationally expensive analysis on cloud
- Machine Learning scientist: train and deploy ML models on the cloud
- Data engineer: build pipelines on the cloud to ingest, transform, and store big data
- Data analyst: access data on the cloud via BI tools
- Cloud architect: solution architect for the cloud
- Cloud engineer: role for anything technical on cloud
- DevOps engineer: Software **D**evelopment + IT **O**perations, IaC
- Security engineer: test and assess security of data on cloud

1.3 Cloud Providers and Case Studies

1.3.1 An overview of providers

1.3.2 Amazon Web Services

1.3.3 Microsoft Azure

1.3.4 Google Cloud